## Document Purpose and Completion Notes

This document records the project development process from problem definition through data collection, preparation, analysis, visualization, interpretation, and final reporting. It is written for the EV Atlas electric-vehicle registration analysis.

Important status note: the project package previously described includes a dashboard, report, presentation, and a draft Tableau workbook. The Tableau workbook must be opened and checked in Tableau before describing its worksheets, dashboards, Story, or publication as fully verified. Do not claim a task was completed unless the student actually performed and checked it.

## Executive Summary

EV Atlas is a data-analysis and visualization project focused on the registered electric-vehicle population in Washington State, USA. The project uses publicly available vehicle-registration information to examine the growth of the registered EV population, the balance between Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs), manufacturer and model patterns, and geographic concentration by county.

The project workflow includes defining the analytical problem, obtaining the source datasets, inspecting their structure, preparing the data, calculating descriptive summaries, creating charts and interactive dashboard views, interpreting the results, and documenting limitations. The reported analysis snapshot contains 300,142 vehicle records and a historical population series with 117 monthly observations from January 2017 to September 2026. These figures should be reconfirmed against the exact downloaded source files used for the submitted analysis.

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# 1. Introduction and Problem Definition

### 1.1 Background

Electric vehicles are an important part of the transition toward lower-emission transport. Public vehicle-registration datasets can help analysts understand how the registered fleet is changing, which vehicle technologies and manufacturers are represented, and where registrations are concentrated. Data visualization makes these patterns easier to explore and communicate.

### 1.2 Problem statement

The project addresses the following problem: how can publicly available electric-vehicle registration data be organized, analysed, and visualized to reveal useful patterns in the Washington State registered EV population? Raw datasets contain many records and fields, so it can be difficult to identify trends and compare categories without a structured analytical process.

### 1.3 Why the problem matters

A time-series view helps describe changes in the registered EV population.

Vehicle-type and manufacturer comparisons help describe the composition of the registered fleet.

County-level summaries show how registrations are distributed geographically.

An interactive dashboard allows a user to investigate the data using filters rather than relying only on static tables.

### 1.4 Project questions

How has the registered EV population changed over the available historical period?

What proportion of the records are BEVs compared with PHEVs?

Which manufacturers and models have the highest registration counts?

Which counties have the largest numbers of registered EVs?

What limitations in the source data should be considered when interpreting the results?

## 2. Project Aim, Objectives and Scope

### 2.1 Aim

To analyse publicly available Washington State electric-vehicle registration data and present the main patterns through clear visualizations and an interactive dashboard.

### 2.2 Objectives

Identify relevant public datasets and document their sources.

Inspect the fields, data types, record counts, and missing values.

Prepare the data for consistent grouping, filtering, and aggregation.

Calculate descriptive counts and proportions for vehicle type, manufacturer, model, county, and historical population.

Create visualizations that answer the project questions.

Build and test an interactive dashboard and, where required, a Tableau Story.

Explain the findings, assumptions, and limitations in a written report and presentation.

### 2.3 Scope

The analysis is descriptive and focuses on the supplied Washington State registration datasets. It describes the records available in the selected data snapshot; it does not estimate total US or global EV sales, revenue, market share, vehicle usage, emissions savings, or consumer preferences.

### 2.4 Intended users

The intended users are students, instructors, and general readers who want to explore summary patterns in the selected public registration data.

## 3. Data Collection and Source Description

### 3.1 Data sources

The project uses two public datasets attributed to the Washington State Department of Licensing (DOL):

### 3.2 Collection procedure

Open the official Washington State open-data portal and locate the two EV datasets by their exact titles.

Review the dataset description, publisher, update date, field definitions, and any usage notes.

Download the original data files in CSV format, retaining an unchanged copy as the raw source.

Record the download date and the dataset update date in the project notes.

Store the source files in a clearly named raw-data folder and use copies for cleaning and analysis.

### 3.3 Source and provenance record

Official portal: https://data.wa.gov/ . Search the portal for “Electric Vehicle Population Data” and “Electric Vehicle Population Size History”. The student should paste the exact dataset-page URLs, access date, and source update date below before submission.

# 4. Data Understanding and Initial Exploration

The purpose of initial exploration is to understand what each row represents, what each field means, and whether the data is suitable for the project questions.

## 4.1 Inspection steps

Load the raw CSV files into a spreadsheet, Python environment, or Tableau Data Source page.

Check the number of rows and columns and inspect a sample of records.

Review column names and data types, especially dates, year, numeric range, and categorical fields.

Check the number of distinct vehicle identifiers, manufacturers, models, counties, and vehicle types.

Summarize missing values and look for inconsistent category labels or unexpected values.

Confirm the time coverage and whether the latest month is a partial or complete reporting period.

## 4.2 Key fields

# 5. Data Cleaning and Preprocessing

Data cleaning prepares the source data for reliable analysis. The student should retain the raw source files and document every transformation applied to the working copies.

## 5.1 Cleaning procedure

Create a separate cleaned-data folder so the original downloaded files remain unchanged.

Standardize column names where needed and preserve a data dictionary linking cleaned names to original fields.

Check vehicle identifiers for duplicates. If a duplicate is found, investigate it and apply a documented rule; do not remove rows solely because several vehicles share the same make or model.

Inspect missing values in county, make, model, model year, electric vehicle type, and electric range.

Keep missing electric-range values as missing or label them as unavailable; do not replace them with zero unless the source definition explicitly says zero is valid.

Standardize category text only where values clearly refer to the same category (for example, removing accidental leading/trailing spaces).

Convert dates to a consistent date format and numeric fields to numeric data types.

Check model years and range values for implausible or out-of-scope entries; flag and investigate rather than silently deleting them.

Check that the historical dataset is sorted by month and has no unintended duplicate month records.

Save the cleaned outputs with clear file names and record the steps taken.

## 5.2 Data quality checks

## 5.3 Cleaning log

# 6. Analysis and Calculation of Measures

## 6.1 Descriptive analysis

After cleaning, descriptive summaries are calculated to answer the project questions. Counts and percentages should be computed from the final cleaned dataset rather than copied from an earlier draft.

## 6.2 Analysis workflow

Create frequency tables for EV type, manufacturer, model, model year, and county.

Calculate percentages using a clearly stated denominator.

Sort rankings in descending order and display only a useful number of categories when charts would otherwise become crowded.

Plot the monthly population series in chronological order.

Compare findings across the two datasets only where the measures and time periods are compatible.

Record the main result, the supporting measure, and the relevant caveat for each project question.

# 7. Visualization and Dashboard Development

## 7.1 Visualization plan

## 7.2 Dashboard design procedure

Choose a clear dashboard title and state the geographic and time scope.

Place key summary indicators near the top, with definitions and units.

Arrange charts in a logical reading order: overall trend, vehicle mix, manufacturers/models, and geography.

Add only useful filters, such as manufacturer, county, vehicle type, and model year.

Use consistent colours for BEV and PHEV across all views.

Add informative titles, axis labels, legends, tooltips, and source notes.

Test each filter and confirm that all relevant charts respond as intended.

Check readability at the screen size used for the presentation.

## 7.3 Tableau development

For the Tableau version, connect to the prepared CSV files, verify field types, create worksheets, assemble the dashboard, and add interactive filters. If the assignment requires a Tableau Story, create sequential story points that explain the trend, vehicle mix/manufacturers, geographic pattern, and data limitations. Save a packaged workbook (.twbx) after testing. A draft workbook is not considered final until it opens without broken data connections and all views have been checked.

## 7.4 Screenshots to insert

[Insert Screenshot 1: Tableau Data Source page showing the connected data and field names.]

[Insert Screenshot 2: Historical EV population line chart.]

[Insert Screenshot 3: BEV vs PHEV comparison.]

[Insert Screenshot 4: Manufacturer and model charts.]

[Insert Screenshot 5: County distribution chart.]

[Insert Screenshot 6: Final dashboard with filters visible.]

[Insert Screenshot 7: Tableau Story points, if required.]

# 8. Results and Interpretation

The following are the reported headline figures from the project materials previously prepared. They are included as a starting point for the results chapter and must be recalculated or confirmed against the exact final cleaned CSV files and the final Tableau workbook before submission.

## 8.1 How to write each finding

For every chart, write three short parts: (1) what the chart shows, (2) the specific value or comparison that supports the statement, and (3) what the result does and does not mean. Avoid unsupported causal explanations.

### 8.2 Example interpretation

The reported dataset contains a larger number of BEV records than PHEV records. BEVs account for approximately 80.6% of the 300,142 reported records, while PHEVs account for approximately 19.4%. This describes the composition of the selected registration dataset; it does not, by itself, explain why consumers selected one vehicle type over another.

### 8.3 Findings table for final verification

## 9. Testing and Validation

Testing checks that the data, calculations, charts, filters, and published output behave as expected.

Reconcile the total record count shown in the analysis with the cleaned dataset.

Check that BEV and PHEV counts add up to the total if these are the only included categories; explain any excluded or unknown categories.

Manually verify selected manufacturer and county counts using a grouped summary or filtered table.

Check that the historical line chart is sorted by date and contains no accidental duplicate months.

Test each dashboard filter individually and in combination.

Check that missing values are not displayed as real zero measurements.

Open the saved packaged workbook on the computer used for submission and confirm that data connections work.

If published to Tableau Public, open the public link in a private/incognito browser window and confirm that the intended dashboard is visible.

### 9.1 Test record

## 10. Conclusion and Recommendations

EV Atlas demonstrates a structured approach to analysing public electric-vehicle registration data. By combining data preparation, descriptive analysis, charts, and an interactive dashboard, the project makes it easier to explore the historical registered population, compare vehicle categories, identify leading manufacturers and models, and examine county-level concentrations.

The reported snapshot suggests that BEVs represent the larger share of the selected records, Tesla has the highest manufacturer count, and King County has the highest county count. These statements must be retained only if they are confirmed in the final analysis.

### 10.1 Recommendations

Keep the data source, download date, and data dictionary with the project so that the analysis can be reproduced.

Refresh the datasets periodically and clearly label the reporting cut-off date.

Use per-capita rates and population denominators for fairer geographic comparisons if suitable county population data is added.

Add charging-station availability or other contextual datasets only if the project scope and data quality allow.

Avoid interpreting registration counts as sales, usage, emissions reductions, or consumer preferences without additional evidence.

## 11. Limitations and Future Scope

The analysis is limited to the geography, fields, and time coverage of the selected Washington State public datasets.

Registration records describe the registered fleet and should not be treated as annual sales or global market share.

Some records may have missing or unavailable values, including electric range and geographic attributes.

County counts are absolute totals; they do not adjust for county population, vehicle ownership, or commuting patterns.

The historical series may be revised by the publisher and the latest period may be incomplete; the source update date must be reported.

The project is descriptive and does not establish causal relationships.

Future work could compare registration counts with county population, charging infrastructure, electricity mix, vehicle prices, or emissions factors, provided that reliable and compatible datasets are obtained.

# 12. References

Washington State Department of Licensing / Washington State Open Data Portal. “Electric Vehicle Population Data.” https://data.wa.gov/ (locate the dataset by its exact title; insert the exact dataset-page URL and access date).

Washington State Department of Licensing / Washington State Open Data Portal. “Electric Vehicle Population Size History.” https://data.wa.gov/ (locate the dataset by its exact title; insert the exact dataset-page URL and access date).

Tableau. Tableau Help and product documentation. https://help.tableau.com/ (cite the specific pages used for connecting data, creating dashboards, stories, and publishing, if applicable).

Any additional libraries, tutorials, or external datasets actually used must be listed here with title, publisher, URL, and access date.

# 13. Appendices and Submission Evidence

## Appendix A – Project folder structure

Keep the project organized so that another person can identify the source data, cleaned data, workbook, report, and evidence.

data/raw/ – unchanged downloaded source files

data/processed/ – cleaned and analysis-ready CSV files

tableau/ – final packaged Tableau workbook (.twbx)

visuals/ – exported charts used in the report

docs/ – project documentation and final report

evidence/ – screenshots, validation notes, and performance evidence (if required)

presentation/ – final presentation slides and video

## Appendix B – Final deliverables checklist

Exact source dataset URLs, download dates, and data update dates.

Raw and cleaned datasets, if required by the assignment.

Final Tableau packaged workbook that opens and has been tested.

Dashboard screenshots and Tableau Story screenshots, if required.

Final presentation slides.

Published Tableau Public URL, only if required and actually published.

Any requested code, test output, or performance recording.

Submission confirmation from the institution portal.

| STUDENT / INSTITUTION DETAILS | TO BE COMPLETED |
|---|---|
| Programme / course | [Enter programme and course] |
| Department / institution | [Enter department and institution] |
| Faculty guide / supervisor | [Enter guide name, if applicable] |
| Academic year / submission date | [Enter academic year and date] |

| Project snapshot (reported values; verify against final data) | Value |
|---|---|
| Vehicle-level records | 300,142 |
| Battery Electric Vehicles (BEV) | 241,840 (about 80.6%) |
| Plug-in Hybrid Electric Vehicles (PHEV) | 58,302 (about 19.4%) |
| Tesla registrations | 122,291 |
| King County registrations | 143,976 |
| Historical series | 117 monthly observations; Jan 2017–Sep 2026 |

| Dataset | Purpose | Main fields / content |
|---|---|---|
| Electric Vehicle Population Data | Vehicle-level analysis | Vehicle identifier, make, model, model year, EV type, county, electric range, and related registration attributes. |
| Electric Vehicle Population Size History | Historical population trend | Monthly observation date and total registered EV population. |

| Provenance item | Student to complete |
|---|---|
| Vehicle dataset page URL | [Paste exact official dataset URL] |
| Historical dataset page URL | [Paste exact official dataset URL] |
| Date downloaded | [DD Month YYYY] |
| Dataset update / snapshot date | [As shown on source page] |
| File names and formats | [Record exact CSV file names] |

| Field / concept | Meaning | Use in analysis |
|---|---|---|
| Vehicle identifier | Identifier for a vehicle record (use only as needed for deduplication). | Record uniqueness and duplicate checks. |
| Make / manufacturer | Vehicle manufacturer name. | Manufacturer ranking and filtering. |
| Model | Vehicle model name. | Model-level comparison. |
| Model year | Vehicle model year. | Year-based distribution. |
| Electric vehicle type | BEV or PHEV classification. | Vehicle-type comparison. |
| County | County associated with the registration record. | Geographic comparison. |
| Electric range | Reported electric driving range, where available. | Range distribution; missing values treated separately. |
| Historical date and population | Month and population total in the history file. | Long-term trend chart. |

| Check | What to verify | Evidence to retain |
|---|---|---|
| Completeness | Missing-value counts by important field. | Missing-value summary or screenshot. |
| Uniqueness | Duplicate vehicle identifiers and duplicate months. | Duplicate-check output. |
| Validity | Correct types, sensible year/range values, valid dates. | Validation notes or code output. |
| Consistency | Consistent labels for EV type, county, make, and model. | Category frequency tables. |
| Reconciliation | Cleaned record count compared with the documented source and transformation rules. | Before/after row-count table. |

| Issue found | Action taken | Rows affected / notes |
|---|---|---|
| [Enter issue] | [Enter exact action] | [Enter count or explanation] |
| [Enter issue] | [Enter exact action] | [Enter count or explanation] |
| [Enter issue] | [Enter exact action] | [Enter count or explanation] |

| Measure | Calculation / interpretation |
|---|---|
| Total registrations | Count of valid vehicle records under the documented record-counting rule. |
| Vehicle-type percentage | (Count for a vehicle type ÷ total valid records) × 100. |
| Manufacturer count | Count of records grouped by manufacturer. |
| Model count | Count of records grouped by model, with make included where needed to distinguish similar model names. |
| County count | Count of records grouped by county. |
| Average / median electric range | Calculate only on valid, non-missing range values and state the number of observations used. |
| Historical change | Compare the population values at two selected dates; explain that population change is not the same as new sales. |

| Question / topic | Recommended visual | What it communicates |
|---|---|---|
| Historical population growth | Line chart | Change in registered EV population over time. |
| BEV vs PHEV | Bar chart or clearly labelled donut chart | Number and share of each EV category. |
| Leading manufacturers | Sorted horizontal bar chart | Manufacturers with the largest registration counts. |
| Leading models | Sorted horizontal bar chart | Most frequently recorded vehicle models. |
| County distribution | Sorted bar chart or map (if geographic fields are valid) | Where registrations are concentrated. |
| Model year distribution | Bar chart | Number of records by model year. |
| Electric range | Histogram or box plot | Distribution of reported range values, excluding or separately identifying missing data. |
| County by EV type | Stacked bar chart | BEV/PHEV composition across selected counties. |

| Result reported in project materials | Reported value | Interpretation / caution |
|---|---|---|
| Total vehicle records | 300,142 | Confirm the final record-counting rule and source snapshot. |
| BEV records | 241,840 (80.6%) | BEVs form the larger share of the reported records. |
| PHEV records | 58,302 (19.4%) | PHEVs form the smaller share of the reported records. |
| Tesla records | 122,291 (40.7%) | Highest manufacturer count in the reported snapshot; not global market share. |
| King County records | 143,976 (about 48.0%) | Largest county count in the reported snapshot; not per-capita adoption. |
| Historical population | 22,424 in Jan 2017; 300,142 in Sep 2026 | Compare like-for-like population measures; this is registered population, not annual sales. |

| Project question | Verified result from final analysis | Chart / worksheet |
|---|---|---|
| Population trend | [Enter verified dates and values] | [Enter worksheet name] |

| Project question | Verified result from final analysis | Chart / worksheet |
|---|---|---|
| BEV vs PHEV | [Enter verified counts and percentages] | [Enter worksheet name] |
| Top manufacturers | [Enter verified top manufacturers] | [Enter worksheet name] |
| Top models | [Enter verified top models] | [Enter worksheet name] |
| Top counties | [Enter verified top counties] | [Enter worksheet name] |
| Range pattern | [Enter valid-range summary and missing count] | [Enter worksheet name] |

| Test | Expected result | Actual result / status |
|---|---|---|
| Workbook opens | No missing data-source or broken-view errors. | [Complete after testing] |
| Dashboard filters | Relevant charts update correctly. | [Complete after testing] |
| Totals and percentages | Match verified grouped calculations. | [Complete after testing] |
| Public link (if required) | Opens for a viewer who is not signed in. | [Complete after testing] |
| Presentation recording | Audio is clear and correct dashboard views are shown. | [Complete after recording] |